

First Normal Form (1NF).

Goal: Make Data Atomic

The initial "Bad Design" features a **Repeating Group** (e.g., "Shirt, Shoes" in a single cell), which makes searching and updating difficult. The **1NF Fix** resolves this by splitting the repeating data into separate rows, ensuring every cell holds only one distinct, atomic value.



SQL Normalization: 1NF (1st Normal Form)



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Order_List



	<u>OrderID</u>	CustomerName	ProductPurchased	Date
1	100	Sam M	Shirt, Shoes	2024-05-15
2	101	Carlos X	Hat	2024-05-16

Repeating Groups!
Bad Design.

Definition:
Eliminate Repeating
Groups. Ensure
Data is Atomic.

Order_Items



	<u>OrderID</u>	CustomerName	<u>ProductID</u>	ProductName	Date
1	100	Sam M	1	Shirt	2024-05-15
2	100	Sam M	2	Shoes	2024-05-15

Atomic Values.
Unique Records.
Good Design.

Second Normal Form (2NF)

Goal: Depend on the Entire Primary Key

Tables with a composite key (e.g., OrderID + ProductID) often suffer from a **Partial Dependency**—where a column like CustomerName relies on only *part* of the key (OrderID), causing data redundancy. The **2NF Fix** resolves this by breaking the data into separate tables, ensuring every column depends entirely on the primary key of its new table.



SQL Normalization: 2NF (2nd Normal Form)



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Order_Detail_Analysis PK (OrderID, ProductID)

	OrderID	ProductID	CustomerName	ProductName	OrderDate
1	100	1	Sam M	Shirt	2024-05-15
		2		Shoes	
2	101	3	Carlos X	Hat	2024-05-16

Partial Dependency! Bad Design.

Products

	ProductID	ProductName
1	1	Shirt
2	2	Shoes
3	3	Hat

Orders

	OrderID	CustomerName	OrderDate
1	100	Sam M	2024-05-15
2	101	Carlos X	2024-05-16

OrderItems

OrderID	ProductID
PK	

Normalized 2NF Tables. Primary Keys defined. Good Design.

Order_Detail_Analysis

	OrderID	ProductID	CustomerName	ProductName
1	100	1	Sam M	Shirt
		2		Shoes
2	101	3	Carlos X	Hat

Third Normal Form (3NF)

Goal: No Dependencies between Non-Key Columns

A **Transitive Dependency** occurs when non-key columns form a chain (e.g., OrderID -> CustomerID -> CustomerName), causing data duplication. The **3NF Fix** breaks this chain by moving customer details into a dedicated Customers table, using a Foreign Key to neatly link them back to the Orders table.



SQL Normalization: 3NF (3rd Normal Form)



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2NF (2nd Normal Form) Table with Transitive Dependency

	<u>OrderID</u>	<u>CustomerID</u>	CustomerName	ProductOrdered	(PK)
1	100	C01	Sam M	Shirt Shoes	
2	101	C02	Carlos X	Hat Shoes	

Transitive Dependency!
CustomerName depends on CustomerID.

Customers

<u>CustomerID</u>	CustomerName	OrderDate
C01	Sam M	2024-05-15
C02	Carlos X	2024-05-16
C03	Mary J	2024-05-16

Orders

	<u>OrderID</u>	<u>CustomerID</u>	ProductOrdered
1	100	C01 (FK)	Shirt
2	101	C02	Hat
3	102	C01	Shoes

Order_Detail_Analysis

	<u>OrderID</u>	<u>ProductID</u>	CustomerName	ProductName
1	100	1 2	Sam M	Shirt Shoes
2	101	3	Carlos X	Hat

Orders

<u>OrderID</u>	<u>CustomerID</u>
PK	FK

Normalized 3NF Tables.
Transitive Dependencies
Removed. Good Design.



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